

Question number	Answer	Additional guidance	Mark
2(a)(i)	- mass of large and small bears calculated (1) 88 / 88.1 / 88.13 (kg)	133.5294117647 and 45.4 88.1294117647058 Bald correct answer = 2 marks Bald masses = 1 mark Bald unrounded = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	An explanation that includes the following points: - because the food that Hank was {stealing / eating} was higher in {kilojoules / energy} (than ants and berries) (1) - Hank was storing energy as {biomass / fat} / hibernating bears were {losing biomass / losing weight / living off their reserves} (1)	ACCEPT calories / fats / carbohydrates eating food throughout the whole year was eating food in the winter when the hibernating bears were not diets of other bears had fewer calories DO NOT ACCEPT NPP	(2)

Question number	Answer	Mark
2(b)(i)	The only correct answer is D <i>A is incorrect because DNA could not be analysed using an electron microscope</i> <i>B is incorrect because DNA could not be analysed using an electron microscope</i> <i>C is incorrect because entomology studies insects</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	A description that includes the following points: • {comparison / differences} of the (pattern /size / number / position) bands} (1) • bands specific to females found on two of the tests but not on Hank's (1) • all (three) bears will have different {banding patterns / (DNA) fingerprints} (1)	ACCEPT use {base sequencing / DNA profiling / genetic fingerprint / gel electrophoresis} DO NOT ACCEPT females will have the same bands ACCEPT there will be (some) differences in the bands for all (three) bears DO NOT ACCEPT all the bands will be different IGNORE they don't have similar bands	(3)